

Research Project Assignment

Conference-Style Introduction and Literature Review

Deadline 13th March, 2026

1 Purpose of the Assignment

The purpose of this project is to introduce you to the process of writing an academic research paper using a professional conference format. In modern scientific practice, conference publications are one of the primary venues for disseminating new research findings. This assignment is designed to help you understand how research problems are presented, how previous work is analyzed, and how scholarly arguments are constructed in academic writing.

You will prepare the **Introduction** and **Literature Review (Related Work)** sections of a research paper. These sections represent the conceptual foundation of any research study and are critical for establishing the motivation, context, and relevance of a research problem.

Through this assignment, you will gain experience in identifying research problems, reviewing scholarly literature, synthesizing information from multiple sources, and presenting technical ideas in a structured and academically appropriate format.

2 Group Formation

You must complete this assignment in small collaborative groups.

- Each group must consist of **two to three students**.
- Individual submissions are not permitted unless explicitly approved by the instructor.
- All group members are expected to contribute to topic selection, literature search, analysis, writing, and formatting.
- The names and your roll numbers of all group members must appear on the first page of the submitted document.

Collaboration reflects the standard practice of academic research where most publications are produced by research teams rather than individual authors.

3 Topic Selection

An Excel file has been provided containing a collection of potential research topics categorized into different research domains and subcategories. The spreadsheet also includes suggested research articles that may serve as starting points for literature exploration.

Each group must select a topic from the provided list by choosing an appropriate category and subcategory. The suggested articles in the spreadsheet should be used as initial references, but you are expected to expand your literature search using academic databases such as IEEE Xplore, ACM Digital Library, Google Scholar, SpringerLink, or ScienceDirect.

The selected topic should represent a meaningful research problem within the relevant field. You should ensure that sufficient scholarly literature exists for conducting a structured literature review.

4 Paper Format

The submitted paper must follow a professional conference formatting style. You are required to use the official IEEE conference template available at:

<https://www.ieee.org/conferences/publishing/templates>

Both **LaTeX** and **Microsoft Word** templates are available. You may use either format.

The IEEE conference template uses a **two-column layout**, which is the standard format used by many international conferences in engineering, computer science, and related disciplines.

If a group chooses to target a specific conference (local or international) with a different formatting style, they may use that conference's official template instead of the generic IEEE template. However, the document must still follow professional conference writing conventions.

5 Structure of the Paper

Although a full conference paper typically contains multiple sections such as methodology, experiments, and conclusions, this assignment focuses on the early conceptual components of research writing.

You must include the following sections:

5.1 Title

The title should clearly describe the research problem or domain addressed in the paper. It should be concise yet informative.

5.2 Abstract

A short abstract (approximately 150–200 words) must be included. The abstract should briefly summarize the research problem, its significance, and the overall focus of the paper.

5.3 Introduction

The introduction should establish the context of the research problem and explain its importance. A well-written introduction typically includes:

- Background information about the research area
- Explanation of the problem being addressed
- Discussion of why the problem is significant
- Limitations or challenges in existing approaches
- The objective or focus of the paper

The introduction should guide the reader from a general overview of the topic toward the specific research problem being discussed.

5.4 Literature Review (Related Work)

The literature review presents an overview of existing research related to the selected topic. You must analyze and synthesize findings from multiple research papers.

A strong literature review should:

- Summarize key contributions of previous studies
- Compare different approaches proposed by researchers
- Identify limitations or gaps in the current literature
- Establish the context for the research problem

The literature review should not simply list research papers. Instead, it should critically discuss how existing studies relate to one another and how they contribute to understanding the research problem.

6 Length of the Paper

Conference papers are typically limited to a small number of pages due to strict publication constraints. Many IEEE conferences allow a maximum of six pages for the main paper.

For this assignment, the expected length is:

2–3 pages in conference format

This page limit includes all elements such as title, abstract, introduction, literature review, figures, and references.

The goal is to simulate the preparation of an early-stage conference paper draft.

7 References and Citation Style

All references must be cited using the **IEEE citation style**. Citations should appear as numbered references within square brackets inside the text.

You must include a minimum of **8 scholarly references**. Preferably, these references should be recent publications from the past five years.

Acceptable sources include:

- Peer-reviewed journal articles
- Conference papers
- Books published by academic publishers
- Reputable digital libraries (IEEE, ACM, Springer, Elsevier)

Sources such as blogs, personal websites, or unverified online content should not be used as primary references.

8 Academic Writing Guidelines

You are expected to follow standard academic writing practices commonly used in scientific publications.

The writing should maintain a formal tone, avoid informal expressions, and present ideas clearly and concisely. Technical terminology should be used appropriately, and arguments should be supported with evidence from cited literature.

Conference papers emphasize clarity and precision. Since page space is limited, writing should be concise while still providing sufficient explanation for readers to understand the research context.

Each paragraph should present a coherent idea and logically connect to the next paragraph.

9 Academic Integrity

All work submitted must be original. Proper citation of sources is mandatory whenever ideas, results, or text from other authors are used.

Direct copying of text from research papers without appropriate quotation and citation constitutes plagiarism and is strictly prohibited.

Submissions may be checked using plagiarism detection tools.

10 Submission Requirements

Each group must submit the following:

- Final paper in PDF format

The file name should follow the format:

`CourseProject_part1_GroupNumber.pdf`

11 Evaluation Criteria

The assignment will be evaluated according to the following criteria:

- Relevance and clarity of the chosen research topic
- Quality of the introduction and problem motivation
- Depth and organization of the literature review
- Use of credible academic references
- Clarity and professionalism of academic writing
- Correct use of conference formatting and citation style

12 Learning Outcomes

Upon completion of this assignment, You should be able to understand the structure of academic conference papers, conduct a structured literature review, synthesize research findings from multiple sources, and present technical ideas in a professional research format.